

COVID-19 ANALYSIS USING POWER BI

ABSTRACT

The COVID-19 pandemic affected millions of people worldwide and generated a huge amount of healthcare-related data. This project focuses on analyzing COVID-19 datasets using Power BI to understand the spread of the virus, recovery rates, active cases, and death statistics across different countries and regions.

The dashboard provides interactive visualizations that help users identify trends, compare country-wise statistics, and analyze confirmed, recovered, and death cases effectively. Various charts, cards, and filters were used to create meaningful insights from the dataset.

The project demonstrates the importance of Business Intelligence tools in healthcare analytics and decision-making.

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1. INTRODUCTION

COVID-19 is one of the most dangerous global pandemics in modern history. Governments, healthcare organizations, and researchers required accurate and timely information to monitor the spread of the disease.

Data analytics and visualization tools play an important role in transforming raw healthcare data into meaningful insights. Power BI is one such business intelligence tool that helps users create interactive dashboards and reports.

This project focuses on analyzing COVID-19 data using Power BI dashboards. The dashboard displays country-wise confirmed cases, active cases, recovered cases, and death statistics using charts and visualizations.

2. OBJECTIVES

The main objectives of this project are:

- To analyze COVID-19 data effectively.
 - To visualize country-wise pandemic statistics.
 - To identify trends in active, recovered, and death cases.
 - To create interactive dashboards using Power BI.
 - To help users understand COVID-19 impacts through data visualization.
 - To improve decision-making using analytics.
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3. PROBLEM STATEMENT

Large amounts of COVID-19 data are generated daily, making it difficult to understand the pandemic situation using raw datasets alone. Traditional reporting methods are not effective for identifying trends and comparisons.

The project aims to solve this problem by creating an interactive Power BI dashboard that converts raw COVID-19 data into visual insights for better understanding and analysis.

4. SCOPE OF THE PROJECT

The scope of the project includes:

- Analysis of COVID-19 statistics.
- Country-wise comparison of cases.
- Interactive filtering and visualization.
- Trend analysis using graphs and charts.
- Business Intelligence reporting using Power BI.

This project can be used by students, researchers, healthcare analysts, and organizations for understanding pandemic data.

5. LITERATURE SURVEY

Several researchers and organizations used data analytics during the COVID-19 pandemic to monitor disease spread and healthcare conditions.

- WHO dashboards provided global case monitoring.
- Researchers used machine learning and analytics for prediction.
- Business Intelligence tools helped governments analyze healthcare data.
- Power BI became popular for creating interactive healthcare dashboards.

This project follows similar analytical approaches using Power BI visualizations.

6. SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB minimum
- Hard Disk: 20 GB free space
- Monitor: Standard display monitor

Software Requirements

- Operating System: Windows 10/11
- Tool Used: Microsoft Power BI Desktop
- Dataset Format: CSV / Excel
- Programming Knowledge: Basic data analytics concepts

7. TOOLS AND TECHNOLOGIES USED

Power BI

Power BI is a business intelligence and data visualization tool developed by Microsoft. It helps in:

- Creating dashboards
- Data transformation
- Interactive reporting
- Real-time analytics
- Visual storytelling

Microsoft Excel / CSV

Used for storing and importing COVID-19 datasets.

Data Visualization Techniques

- Bar Charts
- Line Charts
- Pie Charts
- Donut Charts
- KPI Cards
- Tables
- Filters and Slicers

8. DATASET DESCRIPTION

The dataset contains COVID-19 statistics collected from different countries and regions.

Important Fields

Field Name	Description
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Country/Region Name	Name of the country
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Confirmed	Total confirmed COVID-19 cases
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Field Name	Description
Active	Active cases
Recovered	Total recovered cases
Deaths	Total death cases
New Cases	Newly reported cases

The dataset was cleaned and transformed before visualization.

9. METHODOLOGY

The project methodology includes the following stages:

Step 1: Data Collection

COVID-19 datasets were collected from online sources in CSV/Excel format.

Step 2: Data Cleaning

- Removed null values
- Corrected inconsistent data
- Formatted columns properly

Step 3: Data Import

The cleaned dataset was imported into Power BI Desktop.

Step 4: Data Modeling

Relationships and measures were created for analysis.

Step 5: Dashboard Development

Charts, KPIs, and visualizations were added to create the final dashboard.

Step 6: Analysis and Interpretation

Insights were generated from the visual reports.

10. DASHBOARD EXPLANATION

The dashboard contains multiple visual components for COVID-19 analysis.

KPI Cards

- Maximum Active Cases
- Maximum Death Cases
- Total Recovered Cases

These cards provide quick insights into important statistics.

Country Filter

A slicer is used to filter data by country or region.

Bar Chart – Total Deaths by Country

This chart compares death counts among countries such as Brazil, Mexico, India, Italy, and France.

Line Chart – Recovered Cases by Country

Displays recovery trends across multiple countries.

Area Chart – New Cases by Country

Shows the number of newly reported cases.

Donut Chart

Compares confirmed, recovered, and death cases.

Table Visualization

Displays detailed country-wise statistics including:

- Confirmed Cases
- New Cases
- Active Cases
- Recovered Cases
- Death Cases

11. DATA VISUALIZATION TECHNIQUES

Bar Chart

Used for comparing values between countries.

Line Chart

Used for identifying trends and patterns.

Donut Chart

Used to show percentage contribution.

KPI Cards

Used for displaying important metrics clearly.

Tables

Used for detailed data representation.

12. RESULTS AND FINDINGS

The project provided several important insights:

- Countries such as Brazil and the US recorded high COVID-19 cases.
- Recovery rates improved in many countries over time.
- Active cases fluctuated depending on pandemic waves.
- Death rates varied between countries.
- Interactive dashboards improved data understanding.

The project successfully transformed raw data into meaningful business intelligence reports.

13. ADVANTAGES

- Easy to understand visual reports
 - Interactive filtering options
 - Fast analysis of large datasets
 - Improved decision-making
 - Effective healthcare data monitoring
 - User-friendly dashboard design
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14. LIMITATIONS

- Dataset updates may not be real-time.
- Accuracy depends on the source dataset.
- Limited predictive analysis.

- Internet-based datasets may contain missing values.
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15. FUTURE ENHANCEMENTS

Future improvements can include:

- Real-time data integration
 - AI-based prediction models
 - Regional-level analysis
 - Mobile dashboard optimization
 - Integration with cloud databases
 - Advanced analytics using machine learning
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16. CONCLUSION

The COVID-19 Analysis Dashboard developed using Power BI successfully visualizes pandemic-related data in an interactive and meaningful manner. The project demonstrates how business intelligence tools can help users understand healthcare statistics efficiently.

The dashboard provides valuable insights into active cases, recoveries, deaths, and confirmed cases across countries. This project highlights the importance of data analytics and visualization in modern healthcare systems.

17. REFERENCES

1. Microsoft Power BI Documentation
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 4. Online Research Articles on COVID-19 Analytics
 5. Data Visualization and Business Intelligence Resources
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